

Tensor Train Decomposition Based 2D-Angle Estimation in Bistatic MIMO Radar under Impulsive Noise

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Abstract—To estimate two-dimensional direction-of-departure and direction-of-arrival in bistatic multiple-input multiple-output radar under impulsive noise, this work proposes a robust method combining tensor train (TT) decomposition and the capped Frobenius norm (CFN). First, the five-dimensional received data is modeled via the TT-format that captures the intrinsic structure among the multidimensional measurements, and establishes a one-to-one correspondence between the TT-cores and the Vandermonde factor matrices. Then, for the first TT-core estimation, the CFN is incorporated into the TT-singular value decomposition (TT-SVD) procedure to suppress outliers. Subsequently, the remaining four TT-cores are obtained using the standard TT-SVD. Finally, the Vandermonde-structured factor matrices are estimated from the corresponding TT-cores, enabling accurate angle estimation in impulsive noise environment. Simulation results demonstrate the effectiveness of the proposed algorithm.

Index Terms—Bistatic MIMO radar, tensor train decomposition, capped Frobenius norm, 2D-DOD, 2D-DOA.

I. INTRODUCTION

In recent years, with the growing application of bistatic multiple-input multiple-output (MIMO) radar in intelligent reflecting surfaces (IRS), unmanned aerial vehicles (UAV), and autonomous driving, the two-dimensional direction-of-departure (2D-DOD) and direction-of-arrival (2D-DOA) estimations have attracted considerable research attention. For instance, [1] investigated 2D-DOD and 2D-DOA estimations for non-line-of-sight targets assisted by IRS. In [2], Bayesian tensor decomposition was applied to estimate 2D-DOD and 2D-DOA for intelligent transportation systems. Wen *et al.* employed the rotational invariance technique to achieve 2D-DOD and 2D-DOA estimations in UAV scenarios [3]. Furthermore, [4] adopted a sparse L-shaped array to enhance the

performance of 2D angle estimation in bistatic electromagnetic vector sensor-MIMO (EMVS-MIMO) radar. A PARAFAC based method was proposed [5] with the sparse planar EMVS-MIMO configuration. Studies [6]–[8] utilized tensor-train decomposition (TTD) to estimate four-dimensional angular parameters in bistatic uniform planar array MIMO radar, while [9] developed a four-stage reconstruction algorithm in sparse planar arrays.

It is worth mentioning that the aforementioned studies formulate the additive noise using Gaussian distribution. However, impulsive noise commonly exists in radar [10] and communication systems [11]. For one-dimensional (1D) DOD and DOA estimation under impulsive noise, several methods were proposed, including l_p -PARAFAC [12], l_p -TTD [13], $l_{p,\epsilon}$ -tensor-ring decomposition [14], tensor Bayesian completion [15], and Gaussian-kernel low-order covariance-based maximum likelihood estimation [16]. Nevertheless, compared to 1D angle estimation, 2D angle estimation is more challenging. Due to the structural change in the received data and increased array dimensionality, existing 1D angle estimation algorithms cannot be directly extended to the 2D case.

Inspired by the capped Frobenius norm (CFN) for impulsive noise suppression [17], [18], this paper leverages CFN and TTD to estimate 2D-DOD and 2D-DOA in the presence of outliers. The suggested algorithm uses TTD to capture the intrinsic structural relationship within the five-dimensional data of bistatic MIMO radar. Then, it exploits CFN-assisted TT-singular value decomposition (TT-SVD) to suppress impulsive noise. Using the trilinear alternating least squares (TALS) and higher-order rank-one decomposition, the steering matrices are obtained from the TT-cores, and then 2D-DOD and 2D-DOA are estimated. Simulation results demonstrate that the proposed approach achieves high accuracy of 2D angle estimation in the presence of impulsive noise.

Notations: Scalar, vector, matrix, and tensor are represented

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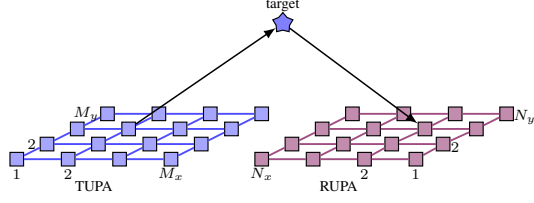


Fig. 1. Bistatic MIMO radar system.

by italic, bold lower-case, bold upper-case, and bold calligraphic letters, respectively. In addition, $(\cdot)^*$, $(\cdot)^T$, $(\cdot)^H$, $(\cdot)^{-1}$, and $(\cdot)^\dagger$ denote conjugate, transpose, conjugate transpose/Hermitian, inverse, and pseudo-inverse, respectively. Besides, $\otimes$ and $\odot$ are the Kronecker product and Khatri-Rao product, respectively. Moreover, $\|\cdot\|_F$, $\angle(\cdot)$, $\arcsin(\cdot)$, and $\arctan(\cdot)$ are Frobenius norm, angle in radians, inverse sine, and inverse tangent, respectively. Furthermore, $\times_n$, $\times_n^p$, and $\mathcal{A}_{(n)}$ denote the n -mode product, tensor contraction product, and mode- n unfolding of $\mathcal{A}$, respectively.

II. SIGNAL MODEL

Consider a bistatic MIMO radar system equipped with uniform planar arrays (UPAs) at the transmitter and the receiver. As shown in Fig. 1, the transmit UPA consists of $M_x \times M_y$ elements, and the receive UPA is composed of $N_x \times N_y$ sensors, with inter-element spacing set to half the wavelength. Suppose there are K far-field targets and their DODs and DOAs are denoted as $\{\theta_{tk}, \phi_{tk}\}_{k=1}^K$ and $\{\theta_{rk}, \phi_{rk}\}_{k=1}^K$, respectively. At time-slot t , the echo data at the receiver is expressed as [7]

$$\mathbf{x}(t) = \sum_{k=1}^K (\mathbf{a}_{tx}(\mu_{tk}) \otimes \mathbf{a}_{ty}(v_{tk}) \otimes \mathbf{a}_{rx}(\mu_{rk}) \otimes \mathbf{a}_{ry}(v_{rk})) \mathbf{s}(t) + \mathbf{n}(t), \quad (1)$$

where $\mu_{tk} = \sin \theta_{tk} \cos \phi_{tk}$, $v_{tk} = \sin \theta_{tk} \sin \phi_{tk}$, $\mu_{rk} = \sin \theta_{rk} \cos \phi_{rk}$, $v_{rk} = \sin \theta_{rk} \sin \phi_{rk}$ denote the direction cosines, $\mathbf{a}_{tx}(\mu_{tk}) = [1, e^{j\pi\mu_{tk}}, \dots, e^{j\pi(M_x-1)\mu_{tk}}]^T \in \mathbb{C}^{M_x \times 1}$, $\mathbf{a}_{ty}(v_{tk}) = [1, e^{j\pi v_{tk}}, \dots, e^{j\pi(M_y-1)v_{tk}}]^T \in \mathbb{C}^{M_y \times 1}$, $\mathbf{a}_{rx}(\mu_{rk}) = [1, e^{j\pi\mu_{rk}}, \dots, e^{j\pi(N_x-1)\mu_{rk}}]^T \in \mathbb{C}^{N_x \times 1}$, $\mathbf{a}_{ry}(v_{rk}) = [1, e^{j\pi v_{rk}}, \dots, e^{j\pi(N_y-1)v_{rk}}]^T \in \mathbb{C}^{N_y \times 1}$ are the steering vectors corresponding to transmit and receive UPAs, while $\mathbf{n}(t) \in \mathbb{C}^{M_x M_y N_x N_y \times 1}$ is additive noise.

To capitalize on the inherent multi-dimensional structure of the received data over L snapshots, the data is formulated in the fifth-order tensor framework

$$\mathcal{X} = \mathcal{I}_{5,K} \times_1 \mathbf{A}_{tx} \times_2 \mathbf{A}_{ty} \times_3 \mathbf{A}_{rx} \times_4 \mathbf{A}_{ry} \times_5 \mathbf{S} + \mathcal{N}, \quad (2)$$

where $\mathbf{A}_{tx} = [\mathbf{a}_{tx}(\mu_{t1}), \mathbf{a}_{tx}(\mu_{t2}), \dots, \mathbf{a}_{tx}(\mu_{tK})] \in \mathbb{C}^{M_x \times K}$, $\mathbf{A}_{ty} = [\mathbf{a}_{ty}(v_{t1}), \mathbf{a}_{ty}(v_{t2}), \dots, \mathbf{a}_{ty}(v_{tK})] \in \mathbb{C}^{M_y \times K}$, $\mathbf{A}_{rx} = [\mathbf{a}_{rx}(\mu_{r1}), \mathbf{a}_{rx}(\mu_{r2}), \dots, \mathbf{a}_{rx}(\mu_{rK})] \in \mathbb{C}^{N_x \times K}$, $\mathbf{A}_{ry} = [\mathbf{a}_{ry}(v_{r1}), \mathbf{a}_{ry}(v_{r2}), \dots, \mathbf{a}_{ry}(v_{rK})] \in \mathbb{C}^{N_y \times K}$, $\mathbf{S} = [\mathbf{s}^T(1), \mathbf{s}^T(2), \dots, \mathbf{s}^T(L)] \in \mathbb{C}^{L \times K}$, and $\mathcal{N} \in \mathbb{C}^{M_x \times M_y \times N_x \times N_y \times L}$.

In [19], TTD is used to capture the internal multilinear relationship within higher-order tensors, thereby preserving rich structural information. Then, $\mathcal{X}$ is decomposed into

$$\mathcal{X} = \mathbf{M}_1 \times_2 \mathcal{M}_2 \times_3 \mathcal{M}_3 \times_4 \mathcal{M}_4 \times_5 \mathbf{M}_5, \quad (3)$$

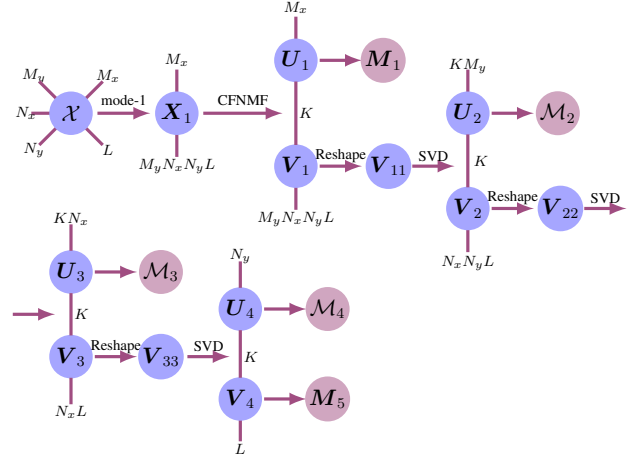


Fig. 2. CFN aided TT-SVD for $\mathcal{X}$.

where $\mathbf{M}_1 \in \mathbb{C}^{M_x \times K}$, $\mathcal{M}_2 \in \mathbb{C}^{K \times M_y \times K}$, $\mathcal{M}_3 \in \mathbb{C}^{K \times N_x \times K}$, $\mathcal{M}_4 \in \mathbb{C}^{K \times N_y \times K}$, and $\mathbf{M}_5 \in \mathbb{C}^{K \times L}$ denote the TT-cores. By setting $\mathcal{I}_{5,K} = \mathbf{I}_K \times_2 \mathcal{I}_{3,K} \times_3 \mathcal{I}_{3,K} \times_4 \mathcal{I}_{3,K} \times_5 \mathbf{I}_K$ and substituting this into (2), we can deduce that $\mathbf{M}_1 = \mathbf{A}_{tx}$, $\mathcal{M}_2 = \mathcal{I}_{3,K} \times_2 \mathbf{A}_{ty}$, $\mathcal{M}_3 = \mathcal{I}_{3,K} \times_2 \mathbf{A}_{rx}$, $\mathcal{M}_4 = \mathcal{I}_{3,K} \times_2 \mathbf{A}_{ry}$, and $\mathbf{M}_5 = \mathbf{S}^T$, indicating that the TT-cores exhibit a one-to-one correspondence with the Vandermonde-structured steering matrix.

III. PROPOSED STRATEGY

A. CFN Aided TT-SVD

To mitigate the effect of impulsive noise, the capped Frobenius norm-based matrix decomposition (CFNMF) is exploited to solve $\mathbf{M}_1$. As illustrated in Fig. 2, the mode-1 unfolding of $\mathcal{X}$ is represented as

$$\mathbf{X}_1 = \mathcal{X}_{(1)}^T = \mathbf{A}_{tx} (\mathbf{S} \odot \mathbf{A}_{ry} \odot \mathbf{A}_{rx} \odot \mathbf{A}_{ty})^T + \mathbf{N} = \mathbf{U}_1 \mathbf{V}_1 + \mathbf{N}. \quad (4)$$

Subsequently, the optimization problem to search for $\mathbf{U}_1$ and $\mathbf{V}_1$ is formulated as

$$\min_{\mathbf{U}_1, \mathbf{V}_1} \left\| \begin{bmatrix} \Re(\mathbf{X}_1 - \mathbf{U}_1 \mathbf{V}_1) \\ \Im(\mathbf{X}_1 - \mathbf{U}_1 \mathbf{V}_1) \end{bmatrix} \right\|_{CF}^2, \quad (5)$$

where $\|\cdot\|_{CF}$ denotes the CFN. For $\mathbf{A} \in \mathbb{C}^{M \times N}$, $\|\mathbf{A}\|_{CF} = \sqrt{\sum_{m=1}^M \sum_{n=1}^N ((\min(\Re(\mathbf{A})_{i,j})^2, \vartheta^2) + (\min(\Im(\mathbf{A})_{i,j})^2, \vartheta^2))}$ [17]. Then, we convert (5) as

$$\begin{aligned} & \min_{\mathbf{U}_1, \mathbf{V}_1, \mathbf{N}} \left\| \begin{bmatrix} \Re(\mathbf{X}_1 - \mathbf{U}_1 \mathbf{V}_1) \\ \Im(\mathbf{X}_1 - \mathbf{U}_1 \mathbf{V}_1) \end{bmatrix} - \begin{bmatrix} \Re(\mathbf{N}) \\ \Im(\mathbf{N}) \end{bmatrix} \right\|_F^2 + h_\vartheta \left(\begin{bmatrix} \Re(\mathbf{N}) \\ \Im(\mathbf{N}) \end{bmatrix} \right) \\ & = \min_{\mathbf{U}_1, \mathbf{V}_1, \mathbf{N}} \ell(\mathbf{U}_1, \mathbf{V}_1, \mathbf{N}) \end{aligned} \quad (6)$$

where $h_\vartheta(\cdot)$ is an entrywise operator. Then, proximal block coordinate descent (PBCD) [20] is adopted as the solver,

resulting in the following iterative procedure

$$\mathbf{N}^{it+1} = \arg \min_{\mathbf{N}} \ell(\mathbf{U}_1^{it}, \mathbf{V}_1^{it}, \mathbf{N}) \quad (7a)$$

$$\mathbf{U}_1^{it+1} = \arg \min_{\mathbf{U}_1} \ell(\mathbf{U}_1, \mathbf{V}_1^{it}, \mathbf{N}^{it+1}) + \varepsilon \left\| \mathbf{U}_1^{it} - \mathbf{U}_1 \right\|_{CF}^2 \quad (7b)$$

$$\mathbf{V}_1^{it+1} = \arg \min_{\mathbf{V}_1} \ell(\mathbf{U}_1^{it+1}, \mathbf{V}_1, \mathbf{N}^{it+1}) + \varepsilon \left\| \mathbf{V}_1^{it} - \mathbf{V}_1 \right\|_{CF}^2, \quad (7c)$$

where $\varepsilon > 0$ is the regularization parameter. The update procedures for $\mathbf{N}^{it+1}$, $\mathbf{U}_1^{it+1}$ and $\mathbf{V}_1^{it+1}$ are similar to those described in [17]. Following the iterative process, $\mathbf{U}_1$, $\mathbf{V}_1$, and $\mathbf{N}$ are effectively estimated as $\mathbf{U}_1 = \mathbf{A}_{tx} \mathbf{\Lambda}_1^{-1}$, $\mathbf{V}_1 = \mathbf{\Lambda}_1 (\mathbf{S} \odot \mathbf{A}_{ry} \odot \mathbf{A}_{rx} \odot \mathbf{A}_{ty})^T$ with $\mathbf{\Lambda}_1$ is the nonsingular change-of-basis matrix. Thus, the TT-core $\mathcal{M}_1$ can be estimated as

$$\mathcal{M}_1 = \mathbf{U}_1 = \mathbf{A}_{tx} \mathbf{\Lambda}_1^{-1}. \quad (8)$$

To solve for $\mathcal{M}_2$, $\mathbf{V}_1$ is first reshaped into $\mathbf{V}_{11} = \text{reshape}(\mathbf{V}_1; KM_y, N_x N_y L) = (\mathbf{A}_{ty} \odot \mathbf{\Lambda}_1) (\mathbf{S} \odot \mathbf{A}_{ry} \odot \mathbf{A}_{rx})^T$. By performing SVD on $\mathbf{V}_{11}$, we obtain

$$\mathbf{U}_2 = (\mathbf{A}_{ty} \odot \mathbf{\Lambda}_1) \mathbf{\Lambda}_2^{-1}, \mathbf{V}_2 = \mathbf{\Lambda}_2 (\mathbf{S} \odot \mathbf{A}_{ry} \odot \mathbf{A}_{rx})^T, \quad (9)$$

where $\mathbf{U}_2$ and $\mathbf{V}_2$ denote the left and right eigenvalue matrix of $\mathbf{V}_{11}$, respectively, and $\mathbf{\Lambda}_2$ is the nonsingular change-of-basis matrix. Then, $\mathcal{M}_2$ is calculated as

$$\mathcal{M}_2 = \mathcal{I}_{3,K} \times_1 \mathbf{\Lambda}_1 \times_2 \mathbf{A}_{ty} \times_3 \mathbf{\Lambda}_2^{-T}. \quad (10)$$

In an analogous procedure to solving for $\mathcal{M}_2$, $\mathcal{M}_3$, $\mathcal{M}_4$ and $\mathcal{M}_5$ are computed as

$$\mathcal{M}_3 = \mathcal{I}_{3,K} \times_1 \mathbf{\Lambda}_2 \times_2 \mathbf{A}_{rx} \times_3 \mathbf{\Lambda}_3^{-T} \quad (11a)$$

$$\mathcal{M}_4 = \mathcal{I}_{3,K} \times_1 \mathbf{\Lambda}_3 \times_2 \mathbf{A}_{ry} \times_3 \mathbf{\Lambda}_4^{-T} \quad (11b)$$

$$\mathcal{M}_5 = \mathbf{\Lambda}_4 \mathbf{S}^T, \quad (11c)$$

where $\mathbf{\Lambda}_3$ and $\mathbf{\Lambda}_4$ are the nonsingular change-of-basis matrices.

B. 2D-DOD and 2D-DOA Estimation

To estimate 2D-DOD and 2D-DOA, it requires extracting $\mathbf{A}_{tx}$, $\mathbf{A}_{ty}$, $\mathbf{A}_{rx}$, and $\mathbf{A}_{ry}$ from $\mathcal{M}_1$, $\mathcal{M}_2$, $\mathcal{M}_3$, $\mathcal{M}_4$, and $\mathcal{M}_5$. First, we combine $\{\mathcal{M}_1, \mathcal{M}_2\}$ and $\{\mathcal{M}_3, \mathcal{M}_4, \mathcal{M}_5\}$ to obtain

$$\mathcal{M}_{12} = \mathcal{M}_1 \times_1^1 \mathcal{M}_2 \in \mathbb{C}^{M_x \times M_y \times K}, \quad (12a)$$

$$\mathcal{M}_{345} = \mathcal{M}_3 \times_3^1 \mathcal{M}_4 \times_4^1 \mathcal{M}_5 \in \mathbb{C}^{K \times N_x \times N_y \times L}, \quad (12b)$$

where $\mathcal{M}_{12} = \mathcal{I}_{3,K} \times_1 \mathbf{A}_{tx} \times_2 \mathbf{A}_{ty} \times_3 \mathbf{\Lambda}_2^{-T}$, $\mathcal{M}_{345} = \mathcal{I}_{4,K} \times_1 \mathbf{\Lambda}_2 \times_2 \mathbf{A}_{rx} \times_3 \mathbf{A}_{ry} \times_4 \mathbf{S}$. Then, trilinear alternating least squares (TALS) is applied to $\mathcal{M}_{12}$ for $\mathbf{A}_{tx}$, $\mathbf{A}_{ty}$ and $\mathbf{\Lambda}_2$ estimation

$$\min_{\mathbf{A}_{tx}, \mathbf{A}_{ty}, \mathbf{\Lambda}_2} \left\| \mathcal{M}_{12} - \mathcal{I}_{3,K} \times_1 \mathbf{A}_{tx} \times_2 \mathbf{A}_{ty} \times_3 \mathbf{\Lambda}_2^{-T} \right\|_F^2. \quad (13)$$

It should be noted that the estimated matrices share the same column scaling and permutation ambiguities, that is, $\hat{\mathbf{A}}_{tx} = \mathbf{A}_{tx} \mathbf{\Pi} \mathbf{\Omega}$, $\hat{\mathbf{A}}_{ty} = \mathbf{A}_{ty} \mathbf{\Pi} \mathbf{\Omega}$, $\hat{\mathbf{\Lambda}}_2 = \mathbf{\Lambda} \mathbf{\Pi} \mathbf{\Omega}$. In particular, the column permutation ambiguity $\mathbf{\Pi}$ affects the pairing of the 2D-DOD and 2D-DOA parameters.

By using $\mathbf{\Pi}$ [8], the column permutation property of $\hat{\mathbf{A}}_{tx}$ is imposed onto $\mathbf{A}_{rx}$ and $\mathbf{A}_{ry}$ in the fourth-order tensor $\mathcal{M}_{345}$. Then, $\hat{\mathcal{M}}_{345}$ is derived as

$$\begin{aligned} \hat{\mathcal{M}}_{345} &= \mathcal{M}_{345} \times_1^1 \hat{\mathbf{\Lambda}}_2^T \\ &= \mathcal{I}_{4,K} \times_1 \mathbf{\Omega} \mathbf{\Pi}^T \times_2 \mathbf{A}_{rx} \times_3 \mathbf{A}_{ry} \times_4 \mathbf{S}. \end{aligned} \quad (14)$$

As the first mode of $\hat{\mathcal{M}}_{345}$ contains $\mathbf{\Pi}$ and $\mathbf{\Omega}$, it lacks angular information. The mode-1 unfolding of $\hat{\mathcal{M}}_{345}$ is

$$\begin{aligned} \hat{\mathbf{M}}^{345} &= \hat{\mathcal{M}}_{345(1)} = (\mathbf{S} \odot \mathbf{A}_{ry} \odot \mathbf{A}_{rx}) (\mathbf{\Omega} \mathbf{\Pi}^T)^T \\ &= \mathbf{S}^\rho \odot \mathbf{A}_{ry}^\rho \odot \mathbf{A}_{rx}^\rho, \end{aligned} \quad (15)$$

where $\mathbf{S}^\rho = \mathbf{S} \mathbf{\Pi} \mathbf{\Omega}$, $\mathbf{A}_{ry}^\rho = \mathbf{A}_{ry} \mathbf{\Pi}$, $\mathbf{A}_{rx}^\rho = \mathbf{A}_{rx} \mathbf{\Pi}$. Leveraging the fact that $\hat{\mathbf{M}}^{345}$ admits a third-order rank-one matrices product structure, the estimates of $\mathbf{S}^\rho$, $\mathbf{A}_{ry}^\rho$, and $\mathbf{A}_{rx}^\rho$ are enabled by employing higher-order rank-one decomposition. First, the following optimization problem is formulated:

$$\min_{\mathbf{S}^\rho, \mathbf{A}_{ry}^\rho, \mathbf{A}_{rx}^\rho} \sum_{k=1}^K \left\| \hat{\mathcal{M}}_k^{345} - \mathbf{s}_k^\rho \circ \mathbf{a}_{ryk}^\rho \circ \mathbf{a}_{rxk}^\rho \right\|_F^2, \quad (16)$$

where $\hat{\mathcal{M}}_k^{345} = \text{reshape}(\hat{\mathbf{M}}^{345}(:, k); L, M_y, M_x)$. The higher-order singular value decomposition (HOSVD) of $\hat{\mathcal{M}}_k^{345}$ is

$$\hat{\mathcal{M}}_k^{345} = \hat{\mathcal{G}}_k \times_1 \mathbf{L}_{k1} \times_2 \mathbf{L}_{k2} \times_3 \mathbf{L}_{k3}, \quad (17)$$

where $\mathbf{L}_{k1} \in \mathbb{C}^{L \times L}$, $\mathbf{L}_{k2} \in \mathbb{C}^{M_y \times M_y}$, $\mathbf{L}_{k3} \in \mathbb{C}^{M_x \times M_x}$ denote unitary matrices, $\hat{\mathcal{G}}_k \in \mathbb{C}^{L \times M_y \times M_x}$ is the core tensor. Thus, $\hat{\mathbf{s}}_k^\rho = \sqrt[3]{\hat{\mathcal{G}}_k(1, 1, 1)} \mathbf{L}_{k1}(:, 1)$, $\hat{\mathbf{a}}_{ryk}^\rho = \sqrt[3]{\hat{\mathcal{G}}_k(1, 1, 1)} \mathbf{L}_{k2}(:, 1)$, $\hat{\mathbf{a}}_{rxk}^\rho = \sqrt[3]{\hat{\mathcal{G}}_k(1, 1, 1)} \mathbf{L}_{k3}(:, 1)$. Then, $\mathbf{S}^\rho$, $\mathbf{A}_{ry}^\rho$, and $\mathbf{A}_{rx}^\rho$ are calculated as

$$\begin{aligned} \hat{\mathbf{S}}^\rho &= [\hat{\mathbf{s}}_1^\rho, \hat{\mathbf{s}}_2^\rho, \dots, \hat{\mathbf{s}}_K^\rho] = \mathbf{S} \mathbf{\Pi} \mathbf{\Omega} \mathbf{\Lambda}_1, \\ \hat{\mathbf{A}}_{ry}^\rho &= [\hat{\mathbf{a}}_{ry1}^\rho, \hat{\mathbf{a}}_{ry2}^\rho, \dots, \hat{\mathbf{a}}_{ryK}^\rho] = \mathbf{A}_{ry} \mathbf{\Pi} \mathbf{\Omega}_1, \\ \hat{\mathbf{A}}_{rx}^\rho &= [\hat{\mathbf{a}}_{rx1}^\rho, \hat{\mathbf{a}}_{rx2}^\rho, \dots, \hat{\mathbf{a}}_{rxK}^\rho] = \mathbf{A}_{rx} \mathbf{\Pi} \mathbf{\Omega}_1. \end{aligned} \quad (18)$$

where $\mathbf{\Omega}_1$ denotes scaling matrix. As derived above, the estimated $\hat{\mathbf{A}}_{tx}$, $\hat{\mathbf{A}}_{ty}$, $\hat{\mathbf{A}}_{ry}^\rho$, and $\hat{\mathbf{A}}_{rx}^\rho$ share an identical $\mathbf{\Pi}$. This ensures that the 2D-DOD and 2D-DOA estimates are automatically paired.

The direction cosines μ_{tk} , v_{tk} , μ_{rk} , and v_{rk} are computed as

$$\hat{\mu}_{tk} = \frac{\sum_{m_x=1}^{M_x-1} \angle \left(\hat{\mathbf{A}}_{tx}(m_x + 1, k) / \hat{\mathbf{A}}_{tx}(m_x, k) \right)}{(M_x - 1)\pi}, \quad (19a)$$

$$\hat{v}_{tk} = \frac{\sum_{m_y=1}^{M_y-1} \angle \left(\hat{\mathbf{A}}_{ty}(m_y + 1, k) / \hat{\mathbf{A}}_{ty}(m_y, k) \right)}{(M_y - 1)\pi}, \quad (19b)$$

$$\hat{\mu}_{rk} = \frac{\sum_{n_x=1}^{N_x-1} \angle \left(\hat{\mathbf{A}}_{rx}^\rho(n_x + 1, k) / \hat{\mathbf{A}}_{rx}^\rho(n_x, k) \right)}{(N_x - 1)\pi}, \quad (19c)$$

$$\hat{v}_{rk} = \frac{\sum_{n_y=1}^{N_y-1} \angle \left(\hat{\mathbf{A}}_{ry}^\rho(n_y + 1, k) / \hat{\mathbf{A}}_{ry}^\rho(n_y, k) \right)}{(N_y - 1)\pi}. \quad (19d)$$

Finally, $\{\theta_{tk}, \phi_{tk}\}_{k=1}^K$ and $\{\theta_{rk}, \phi_{rk}\}_{k=1}^K$ are obtained:

$$\begin{aligned}\hat{\theta}_{tk} &= \arcsin\left(\sqrt{\hat{\mu}_{tk}^2 + \hat{v}_{tk}^2}\right), \hat{\phi}_{tk} = \arctan\left(\frac{\hat{v}_{tk}}{\hat{\mu}_{tk}}\right), \\ \hat{\theta}_{rk} &= \arcsin\left(\sqrt{\hat{\mu}_{rk}^2 + \hat{v}_{rk}^2}\right), \hat{\phi}_{rk} = \arctan\left(\frac{\hat{v}_{rk}}{\hat{\mu}_{rk}}\right).\end{aligned}\quad (20)$$

C. Complexity Analysis

Herein, the complexity analysis is based on $\mathcal{X}$. For the update of M_1, M_2, M_3, M_4 , and M_5 , the complexity is dominated by the calculation of CFN matrix decomposition and three SVDs, resulting in $\mathcal{O}(\kappa_{it}(M_x M_y N_x N_y L(K + K^2)) + K^2(N_x N_y L + N_y L + L))$, where κ_{it} is the iteration number. In addition, the complexity to estimate A_{tx} and A_{ty} by (13) is $\mathcal{O}(\hat{\kappa}_{it}(M_x M_y K^2))$. For estimating A_{rx} , A_{ry} , and S by (16), the complexity is $\mathcal{O}(N_x N_y K L)$. As a result, the total computational complexity is $\mathcal{O}(\kappa_{it}(M_x M_y N_x N_y L K^2) + K^2 N_x N_y L + \hat{\kappa}_{it}(M_x M_y K^2))$. Moreover, we provide the computational complexities of the comparison methods, namely, PARAFAC [5], TTD [8], JIRAFE [19], and FLOM [21]. Their computational complexities are $\mathcal{O}(\hat{\kappa}_{it}(M_x M_y N_x N_y L K))$, $\mathcal{O}(K^2(M_y N_x N_y L) + \hat{\kappa}_{it}(K^3 M_y))$, $\mathcal{O}(K^2(M_y N_x N_y L) + \hat{\kappa}_{it}(K^2 N_y L))$, and $\mathcal{O}(L(M_x M_y N_x N_y)^2 + (M_x M_y N_x N_y)^3)$, respectively.

IV. SIMULATION RESULTS

The element numbers in the transmit and receive UPAs are set as $M_x = M_y = 3$ and $N_x = N_y = 4$, respectively. Three targets are considered, while their 2D-DODs and 2D-DOAs are $(\theta_{t1}, \phi_{t1}) = (20^\circ, 15^\circ)$, $(\theta_{t2}, \phi_{t2}) = (30^\circ, 25^\circ)$, $(\theta_{t3}, \phi_{t3}) = (40^\circ, 35^\circ)$, $(\theta_{r1}, \phi_{r1}) = (16^\circ, 21^\circ)$, $(\theta_{r2}, \phi_{r2}) = (24^\circ, 32^\circ)$, and $(\theta_{r3}, \phi_{r3}) = (38^\circ, 55^\circ)$. The proposed method is compared with PARAFAC [5], TTD [8], JIRAFE [19], and FLOM [21]. Additionally, the Cramér–Rao Bound (CRB) curves are provided as performance lower bounds under both Gaussian noise (GCRB) [22] and impulsive noise (ICRB) [23]. The impulsive noise is modeled as Gaussian mixture noise, whose probability density function is given by $p_v(v) = \frac{c_1}{\pi\sigma_1^2} \exp\left(-\frac{|v|^2}{\sigma_1^2}\right) + \frac{c_2}{\pi\sigma_2^2} \exp\left(-\frac{|v|^2}{\sigma_2^2}\right)$, $c_1 + c_2 = 1$, $\sigma_2^2 \gg \sigma_1^2$. In the following simulation, c_2 is set to 0.1, $\sigma_2^2 = 100\sigma_1^2$. Besides, the signal-to-noise ratio (SNR) and the averaged root mean square error (RMSE) are defined as $\text{SNR} = 10\log_{10}\left(\frac{\|I_{5,K} \times 1 A_{tx} \times 2 A_{ty} \times 3 A_{rx} \times 4 A_{ry} \times 5 S\|_F^2}{M_x M_y N_x N_y L(c_1 \sigma_1^2 + c_2 \sigma_2^2)}\right)$ and $\text{RMSE} = \sqrt{\frac{1}{200K} \sum_{i=1}^{200} (\|\vartheta - \vartheta_i\|_F^2)}$, respectively, where ϑ is the true angle, while ϑ_i denotes the estimated angle in the i th Monte Carlo trials.

Fig. 3(a) shows the RMSE versus SNR with the snapshot number being 50. It is observed that under impulsive noise, JIRAFE and PARAFAC exhibit performance degradation. This is because significant errors in the estimated array steering matrix arise from JIRAFE, which arise from the use of TALS and Khatri-Rao factorization (KRF) when solving the Vandermonde factor matrices after TT-SVD. Although TTD demonstrates relatively robust estimation performance, the proposed algorithm achieves the highest accuracy. This is

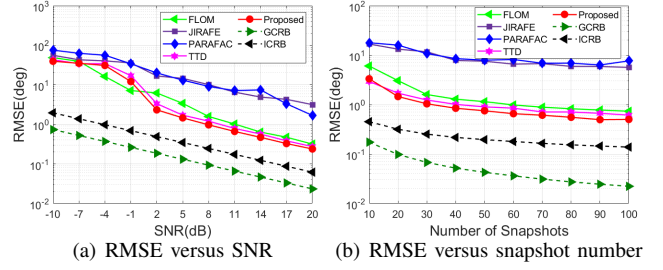


Fig. 3. RMSE versus SNR and snapshots number.

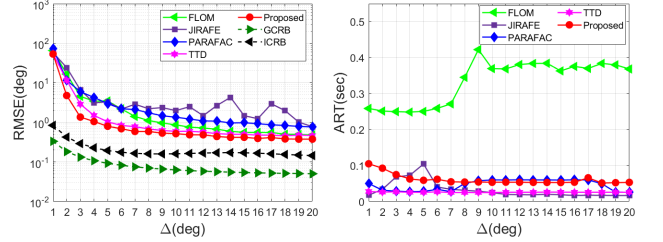


Fig. 4. RMSE comparison of different methods.

because that the approach in [8] applies four sequential TT-SVD steps, whereas the proposed method replaces the first TT-SVD with the robust matrix factorization. Leveraging CFN, this substitution effectively suppresses impulsive noise. Fig. 3(b) shows the RMSE versus snapshot number at SNR = 10dB. Overall, the results validate the effectiveness of the proposed method, demonstrating its superior performance in 2D-DOD and 2D-DOA estimation with impulsive noise.

Fig. 4(a) depicts the RMSE as a function of the angular separation Δ for various algorithms. The 2D-DOD and 2D-DOA pairs for the two targets are set as $(\theta_{t1}, \phi_{t1}) = (20^\circ, 15^\circ)$, $(\theta_{t2}, \phi_{t2}) = (20^\circ + \Delta, 15^\circ + \Delta)$, $(\theta_{r1}, \phi_{r1}) = (16^\circ, 21^\circ)$, and $(\theta_{r2}, \phi_{r2}) = (16^\circ + \Delta, 21^\circ + \Delta)$, respectively. Herein, Δ varies from 1° to 20° with the snapshot number being 50 and SNR being 10dB. It is seen that the proposed algorithm maintains accurate 2D-DOD and 2D-DOA estimation even under small angular separations. In comparison, JIRAFE yields larger estimation errors and exhibits pronounced fluctuations. This validates that the combination of TALS and KRF used in JIRAFE is not well-suited for resolving the Vandermonde-factor matrices. Fig. 4(b) presents the average running time (ART) of different methods. Because the proposed algorithm requires multiple iterations, it has longer ARTs than TTD and JIRAFE.

V. CONCLUSION

In this paper, we have combined CFN and TTD to suppress impulsive noise. The advantages of our algorithm are that the TTD characterizes the intrinsic relationships within the five-dimensional data of bistatic MIMO radar, and the CFN-assisted TT-SVD decomposition effectively mitigates the impact of impulsive noise. Simulation results demonstrate that our method achieves superior performance in 2D-DOD and 2D-DOA estimations under impulsive noise.

REFERENCES

- [1] X. Xue, F. Wen and H. Wang, "Two-dimensional estimation method for bistatic MIMO radar assisted by intelligent reflecting surfaces," in *Proc. IEEE 34th Wireless Opt. Commun. Conf.*, Taipa, Macao, 2025, pp. 260-264.
- [2] W. Yu, J. Du, Y. Chen, L. Jin, X. Li and C. Yuen, "Vehicle localization based on bayesian tensor decomposition in intelligent transportation systems," *IEEE Trans. Intell. Transp. Syst.*, vol. 26, no. 10, pp. 16204-16221, Oct. 2025
- [3] F. Wen, J. Shi, G. Gui, H. Gacanin and O. A. Dobre, "3-D positioning method for anonymous UAV based on bistatic polarized MIMO radar," *IEEE Internet Things J.*, vol. 10, no. 1, pp. 815-827, 1 Jan.1, 2023
- [4] F. Wen, J. Shi, J. He and T. -K. Truong, "2D-DOD and 2D-DOA estimation using sparse L-shaped EMVS-MIMO radar," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 59, no. 2, pp. 2077-2084, April 2023
- [5] X. Wang, Y. Guo, F. Wen, J. He and T. -K. Truong, "EMVS-MIMO radar with sparse Rx geometry: tensor modeling and 2-D direction finding," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 59, no. 6, pp. 8062-8075, Dec. 2023
- [6] Q. Xie, F. Wen, X. Xie, Z. Wang and X. Pan, "Coarray tensor train aided target localization for bistatic MIMO radar," *IEEE Signal Process. Lett.*, vol. 32, pp. 46-50, 2025
- [7] Q. Xie, Z. Wang, F. Wen, J. He and T. -K. Truong, "Coarray tensor train decomposition for bistatic MIMO radar with uniform planar array," *IEEE Trans. Antennas Propag.*, vol. 73, no. 8, pp. 5310-5323, Aug. 2025
- [8] Q. Xie, J. Shi, F. Wen, and Z. Zheng, "Higher-order tensor decomposition for 2D-DOD and 2D-DOA estimation in bistatic MIMO radar," *Signal Process.*, vol. 238, Jan. 2026
- [9] Y. Yue et al., "Four-stage reconstruction-based direction finding for sparse bistatic MIMO radar," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 61, no. 5, pp. 15077-15086, Oct. 2025
- [10] Z. Liu, X. P. Li, and H. C. So, " ℓ_0 -norm minimization based robust matrix completion approach for MIMO radar target localization," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 59, no. 5, pp. 6759-6770, Oct. 2023.
- [11] Z. Liu, H. C. So, X. P. Li, L. Zhang, and Z. Y. Wang, "Robust and energy efficient sparse-coded OFDM-DCSK system via matrix recovery," *IEEE Trans. Commun.*, vol. 71, no. 8, pp. 4839-4850, Aug. 2023.
- [12] X. Lin, L. Huang and W. Sun, " l_p -PARAFAC for joint DOD and DOA estimation in bistatic MIMO radar," in *Proc. 2016 CIE Int. Conf. Radar*, Guangzhou, China, pp. 1-5, 2016.
- [13] Q. Liu, X. Li, H. Cao and Y. Wu, "From simulated to visual data: A robust low-rank tensor completion approach using l_p -regression for outlier resistance," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 6, pp. 3462-3474, June 2022
- [14] X. P. Li and H. C. So, "Robust low-rank tensor completion based on tensor ring rank via $l_{p,\epsilon}$ -norm," *IEEE Trans. Signal Process.*, vol. 69, pp. 3685-3698, 2021
- [15] Y. Song, J. Zhang, J. Chen, G. Tu, and J. Li, "Outlier-resistant bayesian tensor completion for angle estimation in bistatic MIMO radar under array element failures," *Signal Process.*, vol. 236, 2025.
- [16] M. Chen, H. Gao, Y. Du, J. Cheng and Y. Zhang, "Direction finding of bistatic MIMO radar in strong impulse noise," *J. Syst. Eng. Electron.*, vol. 35, no. 4, pp. 888-898, August 2024.
- [17] X. P. Li, Z. Liu, Z. -L. Shi and H. C. So, "MUSIC with capped Frobenius norm: efficient robust direction-of-arrival estimator," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 59, no. 6, pp. 8090-8103, Dec. 2023.
- [18] X. P. Li, Z. -Y. Wang, Z. -L. Shi, H. C. So and N. D. Sidiropoulos, "Robust tensor completion via capped Frobenius norm," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 7, pp. 9700-9712, July 2024
- [19] Y. Zniyed, R. Boyer, A. L.F. de Almeida, and G.d Favier, "Tensor train representation of MIMO channels using the JIRAFE method," *Signal Process.*, vol. 171, pp.107479, 2020.
- [20] X.-P. Li, Z.-L. Shi, Y.-H. Xiao, L. Huang, and H. C. So, "Robust single-bit direction-of-arrival estimation with magnitude recovery," *IEEE J. Sel. Top. Signal Process.*, vol. 19, no. 6, pp. 934-958, Aug. 2025.
- [21] T. H Liu, and J. M. Mendel, "A subspace-based direction finding algorithm using fractional lower order statistics," *IEEE Trans. Signal Process.*, vol. 49, no. 8, pp. 1605-1613, Aug. 2001
- [22] P. Stoica and A. Nehorai, "Performance study of conditional and unconditional direction-of-arrival estimation," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. 38, no. 10, pp. 1783-1795, Oct. 1990
- [23] R. J. Kozick and B. M. Sadler, "Maximum-likelihood array processing in non-Gaussian noise with Gaussian mixtures," *IEEE Trans. Signal Process.*, vol. 48, no. 12, pp. 3520-3535, Dec. 2000.